

Invited Review

A survey of data envelopment analysis in energy and environmental studies

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Abstract

Data envelopment analysis has gained great popularity in energy and environmental (E&E) modeling in recent years. In this paper, we present a literature survey on the application of data envelopment analysis (DEA) to E&E studies. We begin with an introduction to the most widely used DEA techniques, which is followed by a classification of 100 publications in this field. The main features observed are summarized. Issues related to the selection of DEA models in E&E studies are discussed.

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Keywords: Data envelopment analysis (DEA); Energy and environmental (E&E) modeling; Efficiency; Benchmarking; Performance measurement

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1. Introduction

Since the 1973/1974 world oil crisis, energy researchers' enthusiasm in formulating and applying analytical/modeling techniques in energy studies has increased tremendously (Loken, 2007). The enthusiasm was further enhanced by the world-wide awareness and concern on environmental issues in the 1980s. A number of modeling techniques have as a result been developed and employed to address complex energy and environmental (E&E) issues. For example, Jebaraj and Iniyar (2006) reviewed different types of models for energy planning and management. Ang and Zhang (2000) listed a total of 124 studies that applied index decomposition analysis techniques to energy demand and gas emissions analysis. The applications of decision analysis in E&E studies have been reviewed by Huang et al. (1995) and updated by Zhou et al. (2006a).

Among the wide spectrum of E&E modeling techniques, data envelopment analysis (DEA), a relatively new non-parametric approach to efficiency evaluation, has also attracted much attention. Along with the wave of deregulation in energy sectors since the late 1980s, DEA has been accepted as a major frontier technique for benchmarking energy sectors in many countries, particularly in the electricity industry (Jamash and Pollitt, 2001; Abbott, 2005). Although DEA could be treated as a particular multiple criteria decision analysis (MCDA) method, it starts with the purpose of evaluating relative efficiency rather than choosing a specific course of action as is normally the case in the traditional decision analysis (Stewart, 1996). It is therefore not surprising that none of the previous literature reviews on MCDA in E&E studies, such as Huang et al. (1995) and Zhou et al. (2006a), has included DEA-related studies. However, with a growing number of studies using DEA in E&E analysis, a survey of this field would be useful and timely. It is the purpose of this study to fill this gap.

In the sections that follow, we shall first introduce the basic DEA methodology. Next, we introduce the most common extensions to basic DEA models based on our survey and the DEA structure.

We then classify a total of 100 studies published from 1983 to 2006 by the methodological aspect, application scheme, and several other relevant attributes. We present the main features observed and findings. Finally, we discuss some issues on the selection of DEA models and the determination of inputs and outputs, which could be helpful to researchers and practitioners interested in applying DEA to study E&E issues.

2. Basic DEA methodology

Built upon the earlier work of Farrell (1957), DEA is a well established methodology to evaluate the relative efficiencies of a set of comparable entities by some specific mathematical programming models. These entities, often called decision making units (DMUs), perform the same function by transforming multiple inputs into multiple outputs. A main advantage of DEA is that it does not require any prior assumptions on the underlying functional relationships between inputs and outputs (Seiford and Thrall, 1990). It is therefore a nonparametric approach. In addition, DEA is a data-driven frontier analysis technique that floats a piecewise linear surface to rest on top of the empirical observations (Cooper et al., 2004).

Since the work by Charnes et al. (1978), DEA has rapidly grown into an exciting and fruitful field, in which operations research and management science (OR/MS) researchers, economists, and experts from various application areas have played their respective roles (Førsund and Sarafoglou, 2002, 2005). For DEA beginners, Ramanathan (2003) provided an excellent introductory material. The more comprehensive DEA expositions can be found in the recent publication by Cooper et al. (2006). In the sections that follow, we shall briefly introduce the basic DEA methodology.

Assume that there are K DMUs, e.g. electricity distribution utilities, to be evaluated that convert N inputs to M outputs. Further assume that DMU_k consumes $x_{nk} \geq 0$ of input n to produce $y_{mk} \geq 0$ of output m and each DMU has at least one positive input and one positive output (Färe et al., 1994b; Cooper et al., 2004). Based on the efficiency concept

in engineering, the efficiency of a DMU, says DMU_o ($o = 1, 2, \dots, K$), can be estimated by the ratio of its virtual output (weighted combination of outputs) to its virtual input (weighted combination of inputs). To avoid the arbitrariness in assigning the weights for inputs and outputs, [Charnes et al. \(1978\)](#) developed an optimization model known as the CCR in ratio form to determine the optimal weights for DMU_o by maximizing its ratio of virtual output to virtual input while keeping the ratios for all the DMUs not more than one. This problem can be further transformed into an equivalent “output maximization” linear programming problem as follows:

$$\begin{aligned}
 \max \quad & \sum_{m=1}^M u_m y_{mo} \\
 \text{s.t.} \quad & \sum_{m=1}^M u_m y_{mk} - \sum_{n=1}^N v_n x_{nk} \leq 0, \\
 & k = 1, 2, \dots, K, \\
 & \sum_{n=1}^N v_n x_{no} = 1, \\
 & u_m, v_n \geq 0, \quad m = 1, 2, \dots, M; \\
 & n = 1, 2, \dots, N.
 \end{aligned} \tag{1}$$

Model (1) is known as the CCR in multiplier form. The efficiency scores of DMU_1 to DMU_K can be derived by solving K such models. Despite the linear form of (1), efficiency score is usually calculated based on its dual problem:

$$\begin{aligned}
 \min \quad & \theta \\
 \text{s.t.} \quad & \sum_{n=1}^N x_{nk} \lambda_k \leq \theta x_{no}, \quad n = 1, 2, \dots, N, \\
 & \sum_{m=1}^M y_{mk} \lambda_k \geq y_{mo}, \quad m = 1, 2, \dots, M, \\
 & \lambda_k \geq 0, \quad k = 1, \dots, K.
 \end{aligned} \tag{2}$$

Model (2) is known as the input-oriented CCR in envelopment form (or the Farrell model), which attempts to proportionally contract DMU_o 's inputs as much as possible while not decreasing its current level of outputs. In economic literature, model (2) may date back to the activity analysis models introduced by [Von Neumann \(1945\)](#) and [Koopmans \(1951\)](#). It has also a close relationship with the input distance function introduced by [Shephard \(1970\)](#). In a similar way, we can also derive the output-oriented CCR in envelopment form if efficiency is initially specified as the ratio of virtual input to virtual output.

Note that the constraint set in model (2) is nicely corresponding to the piecewise linear production technology that exhibits constant returns to scale (CRS) and has strong disposable inputs and outputs:

$$T = \left\{ (\mathbf{x}, \mathbf{y}) : \begin{aligned} & \sum_{k=1}^K z_k x_{nk} \leq x_n, \quad n = 1, 2, \dots, N, \\ & \sum_{k=1}^K z_k y_{mk} \geq y_m, \quad m = 1, 2, \dots, M, \\ & z_k \geq 0, \quad k = 1, 2, \dots, K \end{aligned} \right\}, \tag{3}$$

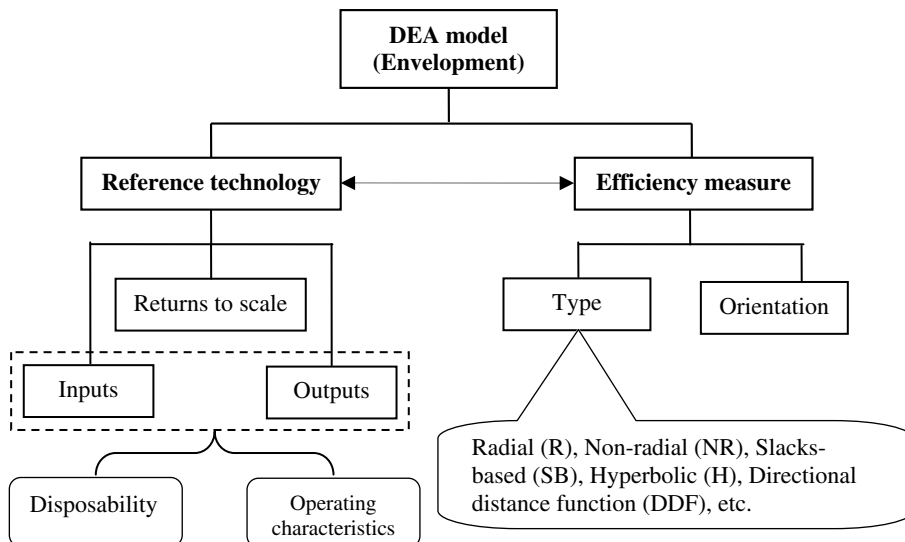


Fig. 1. The general structure of a DEA model (envelopment form).

where $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_m)$ are respectively the vectors of inputs and outputs. Here we call T the reference technology that consists of all the feasible combinations of inputs and outputs.

According to (2) and (3), we may break a DEA model down into two parts: the efficiency measure and the reference technology. A DEA model is fully characterized by its reference technology and efficiency measure. Furthermore, the reference technology can be characterized by the type of returns to scale (RTS), and the disposability and operating characteristics of inputs and outputs. The efficiency measure will be determined by its type and orientation. Fig. 1 shows the general structure of a DEA model as well as the most widely used efficiency measures in E&E studies, which will be discussed in the next section.

3. Extensions to basic DEA models

As was described by Ramanathan (2003) and Cooper et al. (2006), a large number of extensions to basic DEA models have appeared in the literature. We shall limit our discussions to the most widely used extensions in E&E studies based on our survey and the general structure of DEA models (see Fig. 1).

3.1. Reference technology

In traditional DEA models including the CCR, the inputs and outputs are assumed to be strongly or freely disposable. In other words, their reference technologies satisfy that if $(\mathbf{x}, \mathbf{y}) \in T$ and $\mathbf{x}' \geq \mathbf{x}$ (or $\mathbf{y}' \leq \mathbf{y}$) then $(\mathbf{x}', \mathbf{y}) \in T$ (or $(\mathbf{x}, \mathbf{y}') \in T$). However, this may not always be true in the real production process. For instance, in a fossil-fuel-fired electricity generation plant the generation of electricity is always accompanied by the production of undesirable outputs such as sulfur dioxide. In such cases, the reduction of undesirable outputs would likely be costly. It is therefore not appropriate to use the strong disposability reference technology.

Many methods have been proposed to incorporate undesirable outputs into DEA models (Scheel, 2001). Generally, these methods can be divided into two groups. One is based on data translation and the utilization of traditional DEA models, e.g. Seiford and Zhu (2002). The other uses the original data but is based on the concept of weak disposability reference technology as proposed by Färe et al. (1989). In the DEA framework, the weak disposability reference technology, also called the environ-

mental DEA technology (Färe and Grosskopf, 2004), can be characterized as follows:

$$T_e = \left\{ (\mathbf{x}, \mathbf{y}, \mathbf{u}) : \begin{aligned} &\sum_{k=1}^K z_k x_{nk} \leq x_n, \quad n = 1, 2, \dots, N \\ &\sum_{k=1}^K z_k y_{mk} \geq y_m, \quad m = 1, 2, \dots, M \\ &\sum_{k=1}^K z_k u_{jk} = u_j, \quad j = 1, 2, \dots, J \\ &z_k \geq 0, \quad k = 1, 2, \dots, K \end{aligned} \right\}, \quad (4)$$

where $\mathbf{u} = (u_1, u_2, \dots, u_J)$ represents the vector of undesirable outputs.

The difference between T and T_e is that in T_e the reduction of only undesirable outputs is impossible but the proportional reduction of both desirable and undesirable outputs is feasible. In addition, T_e also satisfies the desirable null-jointness property, i.e. if $(\mathbf{x}, \mathbf{y}, \mathbf{u}) \in T$ and $\mathbf{u} = 0$, then $\mathbf{y} = 0$ (Färe et al., 2005). It implies that the only way to eliminate all the undesirable outputs is to end the production process. Therefore, T_e should be a better representation of the real production process when both desirable and undesirable outputs are simultaneously produced. It has been widely applied to such E&E studies as estimating productivity with pollutants considered and modeling environmental performance. See, for example, Färe et al. (1996, 2001, 2004), Chung et al. (1997), Boyd and McClelland (1999), Boyd et al. (2002), Zaim (2004), Arcelus and Arocena (2005), Picazo-Tadeo et al. (2005) and Zhou et al. (2006b, 2007, in press).

Although the derivation of the environmental DEA technology is based on the weak disposability of outputs, similar idea can be generalized to the case of inputs (Färe et al., 1994b). This generalization is particularly useful when DMUs consume such inputs as carbon dioxide (Oude Lansink and Bezlepkin, 2003; Oude Lansink and Silva, 2003).

In addition to the disposability property of inputs and outputs, their operating characteristics, i.e. whether there exist non-discretionary or categorical or environmental variables, sometimes will also play an important role in characterizing the form of a reference technology. Two well-known examples are the DEA models with non-discretionary and categorical variables as formulated by Banker and Morey (1986a,b). In E&E studies, such kinds of

models can be used to measure the efficiency of energy utilities when environmental regulations are imposed or there are external non-controllable factors, e.g. Korhonen and Syrjänen (2003), Agrell and Bogetoft (2005) and Hattori et al. (2005).

Another major stamp of the reference technology is its property on returns to scale (RTS). It is known that the reference technology T for the CCR model exhibits constant returns to scale (CRS). If an additional constraint $\sum_{k=1}^K \lambda_k = 1$ is appended to T , the resulting reference technology will permit the existence of variant returns to scale (VRS) and the CCR model (2) becomes the classical BCC model (Banker et al., 1984).¹ In addition to CRS and VRS, non-increasing returns to scale (NIRS) reference technology, which can be derived by appending $\sum_{k=1}^K \lambda_k \leq 1$ to T , is also very useful because it is helpful to investigate the RTS properties of DMUs (Ramanathan, 2003). Although previous discussions are based on the strong disposability reference technology, various RTS conditions can also be integrated with the weak disposability reference technology in appropriate ways (Färe et al., 1994b; Zhou et al., in press).

3.2. Efficiency measures

Efficiency measure will completely determine a DEA model once the reference technology is given (see Fig. 1). From the orientation point of view, efficiency measures used in E&E studies mainly consist of inputs, outputs and undesirable outputs oriented measures. In the case of the type, many different efficiency measures have been proposed with their respective advantages. Here we shall only introduce those which have been widely adopted in E&E studies.

Radial efficiency measure, which adjusts inputs or outputs proportionally, is probably the most widely used in DEA models. By combining radial efficiency measure with various reference technologies, we can obtain various DEA models including the CCR and BCC. If T_e is used and radial efficiency measure for adjusting undesirable outputs is adopted, we will get such a model as $\min\{\theta: (\mathbf{x}_o, \mathbf{y}_o, \theta \mathbf{u}_o) \in T_e\}$ that can be used to measure

the environmental performance of DMU_{*o*}, e.g. Tyteca (1996, 1997) and Färe et al. (2004).

Non-radial efficiency measure allows for the non-proportional adjustment of different inputs/outputs. It usually has a higher discriminating power than radial efficiency measure in comparing DMUs. A well-known non-radial efficiency measure is the Russell efficiency measure $\min\{\frac{1}{N} \sum_{n=1}^N \theta_n : (\mathbf{x}_o \theta, \mathbf{y}_o) \in T\}$ where θ is a diagonal matrix consisting of θ_1 to θ_n (Färe et al., 1994b). If the weights for θ_n ($n = 1, 2, \dots, n$) are given, the weighted non-radial efficiency measure reflecting the preference structure of decision makers can also be obtained (Zhu, 1996).

Slacks-based efficiency measure is constructed directly from the slacks in inputs and outputs. There are various slacks-based efficiency measures, e.g. the additive measure and the Tone's measure (Cooper et al., 2006). Since slacks-based efficiency measure can identify all the economic inefficiencies, its discriminating power is relatively high.

Hyperbolic efficiency measure, also called graph measure, attempts to simultaneously reduce inputs and expand outputs at the same rate (Färe et al., 1994b). Technically, it can be characterized as $\min\{\theta: (\theta \mathbf{x}_o, \mathbf{y}_o / \theta) \in T\}$ if T is the reference technology used. This measure is particular useful when there are both desirable and undesirable outputs.

Directional distance function (DDF) efficiency measure allows us to simultaneously expand desirable outputs and reduce inputs and/or undesirable outputs based on a given direction vector (Chung et al., 1997). It represents a more general concept since traditional radial efficiency measure is a special case for it (Färe and Grosskopf, 2004).

3.3. Nonparametric Malmquist productivity index

In the foregoing the use of DEA is restricted to cross-sectional analysis, i.e. multilateral comparisons among different DMUs at the same point in time. However, in the case of energy sectors, there is generally a great interest in investigating their productivity change over time.

The nonparametric Malmquist productivity index (MPI) is such a formal time-series analysis method for conducting performance comparisons of DMUs over time by solving some DEA-type models (Malmquist, 1953; Caves et al., 1982; Färe et al., 1994a). Although MPI is defined based on the concept of distance function, it can also be directly represented by DEA efficiency measures. Assume that $\theta^t(\mathbf{x}_o^t, \mathbf{y}_o^t)$ and $\theta^{t+1}(\mathbf{x}_o^t, \mathbf{y}_o^t)$ are the

¹ It is worth pointing out that the so called BCC model (in the case of single output) was first introduced by Afriat (1972) and later discussed by Färe et al. (1983). More discussions on the history of DEA developments can be found in Førsund and Srafolglou (2002, 2005).

input-oriented efficiency measures of DMU_o based on its inputs and outputs at period t for the reference technology at t and $t + 1$. Further assume that $\theta^t(\mathbf{x}_o^{t+1}, \mathbf{y}_o^{t+1})$ and $\theta^{t+1}(\mathbf{x}_o^{t+1}, \mathbf{y}_o^{t+1})$ are the input-oriented efficiency measures of DMU_o based on its inputs and outputs at period $t + 1$ for the reference technology at t and $t + 1$. The output-oriented MPI can be defined as

$$MPI_o = \left[\frac{\theta^t(\mathbf{x}_o^{t+1}, \mathbf{y}_o^{t+1})}{\theta^t(\mathbf{x}_o^t, \mathbf{y}_o^t)} \frac{\theta^{t+1}(\mathbf{x}_o^{t+1}, \mathbf{y}_o^{t+1})}{\theta^{t+1}(\mathbf{x}_o^t, \mathbf{y}_o^t)} \right]^{1/2}. \quad (5)$$

We can then use MPI_o to measure the productivity change of DMU_o over time. $MPI_o > 1$, $MPI_o = 1$ and $MPI_o < 1$ respectively indicate that the productivity of DMU_o has improved, remained unchanged, and deteriorated from period t to $t + 1$.

Following Färe et al. (1994a), MPI_o can also be written as

$$MPI_o = \left[\frac{\theta^t(\mathbf{x}_o^t, \mathbf{y}_o^t)}{\theta^{t+1}(\mathbf{x}_o^t, \mathbf{y}_o^t)} \frac{\theta^t(\mathbf{x}_o^{t+1}, \mathbf{y}_o^{t+1})}{\theta^{t+1}(\mathbf{x}_o^{t+1}, \mathbf{y}_o^{t+1})} \right]^{1/2} \times \frac{\theta^{t+1}(\mathbf{x}_o^{t+1}, \mathbf{y}_o^{t+1})}{\theta^t(\mathbf{x}_o^t, \mathbf{y}_o^t)}. \quad (6)$$

In such a way, the productivity change can be decomposed into two parts, namely technological change and efficiency change. The technological change component, i.e. the terms enclosed by the square brackets, reflects the shift in the best practice frontier from t to $t + 1$. The efficiency change component, i.e. the terms outside the brackets, measures the change in relative efficiency over time.

Other aspects of MPI have been explored in a variety of studies. For more details, see the excellent MPI survey study by Färe et al. (1998).

3.4. Miscellaneous

In addition to those covered in Sections 3.1–3.3, there are many other theoretical extensions to basic DEA models. Among these extensions, some are based on the envelopment version of DEA models, e.g. the non-separating approach proposed by Scheel (2001) to dealing with undesirable outputs, while others favour DEA models in the multiplier form. In the latter line, the incorporation of weight restrictions into DEA, e.g. the Absolute Weights Restrictions (AWR) approach initiated by Dyson and Thanassoulis (1988), should represent one of the most significant developments in DEA (Allen et al., 1997). The DEA models with weight restric-

tions, which are dual to the slacks-based DEA models proposed by Tone (2001), not only have higher discriminating power in performance comparisons but also are important for understanding the nature of the reference technology (Podinovski, 1999, 2001; Dyson et al., 2001). More discussions on these developments can be found in Allen et al. (1997) and Cooper et al. (2006).

To a certain extent, the popularity of DEA in empirical applications owes to the availability of many specialized DEA software packages. The study by Ramanathan (2003) has listed a collection of useful internet sites and popular DEA software. In E&E studies, two widely used free software packages are the DEAP developed by Coelli (1996) and the EMS developed by Scheel (2000).

4. Main features and findings of past studies

The 100 studies listed in Table 1 have been collected primarily from major OR/MS journals such as *European Journal of Operational Research*, as well as some major E&E and economics journals. They are classified according to the following attributes: type of study, country/region, methodological aspect, and application scheme. The third column of Table 1 shows that the 100 studies cover a wide spectrum of countries. To study possible changes over time, we divide the time frame into three 8-year periods, 1983–1990, 1991–1998 and 1999–2006. The total numbers of publications in these three periods are respectively 7, 21, and 72, which show significant increases over time. In the following sections, we present our main findings on the features of past studies in terms of application scheme and methodological aspect.

4.1. Application scheme

Application scheme refers to the main application issue studied, which is shown in the last column of Table 1. If the purpose of a DEA study is to measure the productive efficiency of a sample of energy utilities, we then use the name of DMUs, e.g. electricity distribution utilities, to represent the application scheme.

It is found that 38% of studies deal with the issues in electricity industry. Prior to 1990, the use of DEA in electricity industry mainly focused on electricity generation plants, e.g. Färe et al. (1983, 1985, 1990). Since the earlier 1990s, DEA has gradually become a popular benchmarking tool for studying

Table 1
Studies of DEA in E&E with their specific features

Publication	Type of study	Country/region	Methodological aspect				Application scheme	
			Reference technology			Efficiency measure		
			Inputs	Outputs	RTS			
Abbott (2006)	A	Australia	SD	SD	C, V	R	Yes	Electricity distribution utilities
Agrell and Bogetoft (2005)	T + A	Denmark	SD	SD + NC	C, V	SB	Yes	District heating plants
Arcelus and Arocena (2005)	A	14 OECD countries	SD	SD, WD	V	R, DDF	No	Productivity estimation with CO ₂ emissions considered
Athanassopoulos et al. (1999)	T + A	UK	SD + B	SD + B	C	NR	No	Electricity generation plants
Bagdadioglu et al. (1996)	A	Turkey	SD	SD	C, V	R	No	Electricity distribution utilities
Barla and Perelman (2005)	A	12 OECD countries	SD	SD	C	R	Yes	Relationship between productivity and SO ₂ emissions
Bevilacqua and Braglia (2002)	A	Italy	SD	SD	C	R	No	Environmental performance measurement
Boyd and McClelland (1999)	T + A	US	SD	SD, WD	C	H	No	Impacts of environmental regulations
Boyd and Pang (2000)	A	US	SD	SD	C	R	No	Energy efficiency study
Boyd et al. (2002)	T + A	US	SD	SD, WD	C	DDF	Yes	Impacts of environmental regulations
Brännlund et al. (1998)	T + A	Sweden	SD	WD	NI	Profit	No	Profit estimation with emissions trading
Byrnes et al. (1984)	T + A	US	SD, WD	SD	C, V, NI	R	No	Coal mines
Byrnes et al. (1988)	T + A	US	SD, WD	SD	C, V, NI	R	No	Coal mines
Callens and Tyteca (1999)	T	–	CCR multiplier form with bad outputs considered				No	Environmental performance measurement
Chauhan et al. (2006)	A	India	SD	SD	C, V	R	No	Energy use efficiency study
Chien et al. (2003)	A	China (Taiwan)	SD	SD	C, V	R	No	Electricity distribution districts
Chitkara (1999)	A	India	SD	SD	C	R	Yes	Electricity generation plants
Chung et al. (1997)	T + A	Sweden	SD	WD	C	DDF	Yes	Productivity estimation with pollutants considered
Claggett and Ferrier (1998)	A	US	SD	SD	C, V	Cost	No	Electricity distribution utilities
Cook and Green (2005)	T + A	–	CCR multiplier form + AR method				No	Electricity generation plants
Criswell and Thompson (1996)	A	US	CCR multiplier form + AR method				No	Comparison of different power systems
Dyckhoff and Allen (2001)	T	–	DEA + multi-attribute value theory				No	Environmental performance measurement

(continued on next page)

Table 1 (continued)

Publication	Type of study	Country/region	Methodological aspect				Application scheme	
			Reference technology			Efficiency measure		
			Inputs	Outputs	RTS			
Edvardsen and Førsund (2003)	T + A	Five European countries	SD	SD	C	R	Yes	Electricity distribution utilities
Färe et al. (1983)	T + A	US	SD,WD	SD	C, V, NI	R	No	Electricity generation plants
Färe et al. (1985)	T + A	US	SD, WD	SD	C, V, NI	R, cost	No	Electricity generation plants
Färe et al. (1986)	T + A	US	SD	SD, WD	V	R	No	Impacts of environmental regulations
Färe et al. (1989)	T + A	US	SD	SD, WD	C	H	No	Impacts of environmental regulations
Färe et al. (1990)	T + A	US	SD	SD	C	R	Yes	Electricity generation plants
Färe et al. (1996)	T + A	US	SD	SD, WD	C	R	No	Environmental performance measurement
Färe et al. (2001)	T + A	US	SD	WD	C	DDF	Yes	Productivity estimation with pollutants considered
Färe et al. (2004)	T + A	17 OECD countries	SD	WD	C	R	No	Environmental performance measurement
Ferrier and Hirschberg (1992)	A	US	SD	SD	C	R	No	Climate control efficiency evaluation
Førsund and Kittelsen (1998)	A	Norway	SD	SD	C	R	Yes	Electricity distribution utilities
Giannakis et al. (2005)	A	UK	SD	SD	C, V	R	Yes	Electricity distribution utilities
Golany et al. (1994)	A	Israel	SD	SD	V	R	No	Electricity generation plants
Goto and Tsutsui (1998)	A	Japan & US	CCR/BCC multiplier form + AR method				Yes	Electricity generation plants
Hattori et al. (2005)	A	Japan & UK	SD	SD + NC	C, V	R	Yes	Electricity distribution utilities
Hawdon (2003)	A	33 countries	SD	SD	C, V	R	No	Gas industry
Haynes et al. (1994)	T	–	CCR multiplier form					Environmental performance measurement
Hernández-Sancho et al. (2000)	T + A	Spain	SD	WD	C	H	No	Impacts of environmental regulations
Hjalmarsson and Veiderpass (1992)	A	Sweden	SD	SD	C	R	Yes	Electricity distribution utilities
Hu and Kao (2007)	T + A	17 APEC economies	SD	SD	C	R, SB	No	Energy efficiency study
Hu and Wang (2006)	T + A	China	SD	SD	C	R, SB	No	Energy efficiency study
Jamasb et al. (2004)	A	US	SD	SD	C	R	No	Electricity distribution utilities
Jamasb and Pollitt (2003)	A	Six European countries	SD	SD	C, V	R	No	Electricity distribution utilities
Kashani (2005a)	A	Norway	SD	SD	C, V	R	Yes	Petroleum industry
Kashani (2005b)	A	UK	SD	SD	C, V	R	Yes	Petroleum industry
Korhonen and Luptacik (2004)	T + A	Europe	CCR extensions with bad outputs considered				No	Environmental performance measurement
Korhonen and Syrjanen (2003)	A	Finland	SD + NC	SD	C, V	R	Yes	Electricity distribution utilities

Table 1 (continued)

Publication	Type of study	Country/region	Methodological aspect				Application scheme	
			Reference technology			Efficiency measure		
			Inputs	Outputs	RTS			
Kulshreshtha and Parikh (2002)	A	India	SD	SD	C	R	Yes	Coal mines
Kumar (2006)	A	41 countries	SD	WD	C	DDF	Yes	Productivity estimation with CO ₂ considered
Lam and Shiu (2001)	A	China	SD	SD	V	R	No	Electricity generation plants
Liang et al. (2004)	T + A	China	Extensions of CCR multiplier form					Environmental performance measurement
Lo et al. (2001)	A	China (Taiwan)	SD	SD	C, V	R	No	Electricity distribution utilities
Miliotis (1992)	A	Greece	SD	SD	C	R	No	Electricity distribution utilities
Munksgaard et al. (2005)	A	Denmark	SD	SD	C, V	R	No	District heating production
Murillo-Zamorano (2005)	A	18 countries	SD	SD	C	R	Yes	Relationship between productivity and energy inputs
Nag (2006)	A	India	SD	SD	C	R	No	Electricity generation plants
Olatubi and Dismukes (2000)	A	US	SD	SD	V	Cost	No	Electricity generation plants
Onut and Soner (2006)	A	Turkey	SD	SD	C	R	No	Energy efficiency study
Oude Lansink and Bezlepkin (2003)	A	The Netherlands	WD + NC	SD	C, V	R	No	CO ₂ and energy efficiency study
Oude Lansink and Silva (2003)	A	The Netherlands	WD + NC	SD	C, V	R	No	CO ₂ and energy efficiency study
Pacudan and de Guzman (2002)	A	Philippines	SD	SD	C, V	R	No	Electricity distribution utilities
Pahwa et al. (2002)	A	US	SD	SD	C	R	No	Electricity distribution utilities
Park and Lesourd (2000)	A	Korea	SD	SD	V	R	No	Electricity generation plants
Pasurka (2006)	T + A	US	SD	WD	C	R	No	Decomposition of air emissions
Picazo-Tadeo et al. (2005)	T + A	Spain	SD	SD, WD	V	DDF	No	impacts of environmental regulations
Pollitt (1996)	A	UK	SD + NC	SD	C, V	R, cost	No	Nuclear electricity generation plants
Pombo and Taborda (2006)	A	Colombia	SD	SD	C, V	R	Yes	Electricity generation plants
Price and Weyman-Jones (1996)	A	UK	SD	SD	C	R	Yes	Gas industry
Raczka (2001)	A	Poland	SD	SD	C, V	R	No	District heating plants
Ramanathan (2000)	A	India	CCR multiplier form				No	Energy efficiency study
Ramanathan (2001)	A	–	CCR multiplier form				No	Comparison of energy supply technologies
Ramanathan (2002)	A	64 countries	SD	SD	C	R	No	Environmental performance measurement
Ramanathan (2005a)	A	17 countries	SD	SD	C, V	R	Yes	Environmental performance measurement
Ramanathan (2005b)	T + A	India	SD	SD	C	SB	No	Projection of energy consumption and CO ₂ emissions

(continued on next page)

Table 1 (continued)

Publication	Type of study	Country/region	Methodological aspect				Application scheme	
			Reference technology			Efficiency measure		
			Inputs	Outputs	RTS			
Resende (2002)	A	Brazil	SD	SD	C, V	R	No	Electricity distribution utilities
Sanhueza et al. (2004)	A	Chile	SD	SD	C	R	No	Electricity distribution utilities
Sarkis and Weinrach (2001)	A	US	CCR multiplier form and its extensions				No	Evaluation of waste treatment technologies
Sueyoshi (1999)	T + A	Japan	SD	SD	V	Cost	No	Electricity generation plants
Sueyoshi (2000)	T + A	Japan	Stochastic extension of CCR multiplier form				No	Petroleum industry
Sueyoshi and Goto (2001)	T + A	Japan	SD	SD		R, SB	Yes	Electricity generation plants
Thakur et al. (2006)	A	India	SD	SD	C, V	R	No	State-owned electric utilities
Thompson et al. (1992)	T + A	US	CCR multiplier form + AR				No	Coal mines
Thompson et al. (1995)	T + A	US	CCR multiplier form + AR				No	Petroleum industry
Thompson et al. (1996)	A	US	CCR multiplier form + AR				No	Petroleum industry
Triantis and Otis (2004)	T	–	Dominance-based DEA extensions				No	Environmental performance measurement
Tyteca (1997)	T + A	US	SD	WD	C	R	No	Environmental performance measurement
Tyteca (1998)	T + A	US	CCR multiplier form with bad outputs considered				No	Environmental performance measurement
Weyman-Jones (1991)	A	UK	SD	SD	C	R	No	Electricity distribution utilities
Yaisawarng and Klein (1994)	T + A	US	SD + NC	WD	C, V, NI	R	No	Electricity generation plants
Yunos and Hawdon (1997)	A	27 developing countries	SD	SD	C, V	R	Yes	Electricity generation plants
Zaim (2004)	T + A	US	SD	WD	C	R	No	Environmental performance measurement
Zaim and Taskin (2000a)	T + A	24 OECD countries	SD	SD, WD	C	H	No	Impacts of environmental regulations
Zaim and Taskin (2000b)	T + A	24 OECD countries	SD	SD, WD	C	H	No	Impacts of environmental regulations
Zhang and Bartels (1998)	T + A	Australia, Sweden and New Zealand	SD	SD	C	R	No	Electricity distribution utilities
Zhou et al. (2006b)	T + A	30 OECD countries	SD	SD, WD	C	R, SB	No	Environmental performance measurement & impacts of environmental regulations
Zhou et al. (in press)	T + A	8 world regions	SD	WD	C, V, NI	R	No	Environmental performance measurement
Zhou et al. (2007)	T + A	26 OECD countries	SD	WD	C	NR	Yes	Environmental performance measurement
Zofio and Prieto (2001)	T + A	14 OECD countries	SD	SD, WD	C	H, SB	No	Environmental performance measurement

T: theory; A: application; SD: strong disposable; WD: weak disposable; NC: non-controllable; B: bounded; AR: assurance region; RTS: returns to scale; C: constant returns to scale; V: variant returns to scale; NI: non-increasing returns to scale; R: radial; NR: non-radial; SB: Slacks-based; H: hyperbolic; DDF: directional distance function; MPI: Malmquist productivity index.

the efficiency of electricity distribution utilities. The study by Weyman-Jones (1991), in which the technical efficiency of the UK electricity distribution industry was studied, is probably the first publication in this line. After that, many studies have appeared in the literature and the study scope has also expanded from a single country case to a cross-country case (also called international benchmarking). Examples of such studies include Hjalmarsson and Veiderpass (1992), Bagdadioglu et al. (1996), Yunos and Hawdon (1997), Førsund and Kittelsen (1998), Edvardsen and Førsund (2003), Jamasb and Pollitt (2003) and Giannakis et al. (2005). Temporally, the shares taken up by the studies on electricity generation plants and the studies on electricity distribution utilities have, respectively, increased from 5% and 7% in 1991–1998 to 10% and 13% in 1999–2006. This could be explained by the electricity sector reforms occurred in many countries since the late 1980s and regulators have chosen appropriate benchmarking techniques such as DEA to quantify the efficiency improvements of electricity utilities.

Modeling environmental performance, which mainly includes environmental performance measurement and estimation of environmental regulation impacts, is another popular application area of DEA in E&E studies. Of the 100 studies, about a quarter deals with this area. Although a few studies were reported before 1990, e.g. Färe et al. (1986, 1989), most of them appeared in the latter two time periods, especially in 1999–2006. It might be the result of the world-wide concern on environmental issues and sustainable development, as well as the ability of DEA in providing a standardized and aggregated environmental performance index.² In some studies, e.g. Tyteca (1997, 1998), Boyd and McClelland (1999), Hernández-Sancho et al. (2000) and Boyd et al. (2002), DEA was used to model environmental performance at firm level. Nevertheless, it seems that there is an increasing tendency of applying DEA to model environmental performance at the macro level. Particularly, regional/national carbon dioxide emissions have been widely studied. See, for example, Zaim and Taskin (2000a,b), Zofio and Prieto (2001), Ramanathan (2002, 2005a), Färe et al. (2004) and Zhou et al. (2006b, in press, 2007). This might be the result of

the growing concern on climate change due to carbon dioxide emissions in recent years.

Energy efficiency measurement and monitoring has evolved as an important topic in E&E studies (Ang, 2006). Recently, the potential of DEA in energy efficiency study has also been widely investigated by researchers. The study by Boyd and Pang (2000) discussed the relationship between productivity and energy intensity, while Ramanathan (2000) applied DEA to study the energy efficiencies of alternative transport modes. More recently, Hu and Wang (2006) and Hu and Kao (2007) developed a total-factor energy efficiency index by using DEA, which provides a useful alternative to the traditional energy efficiency indicators such as the aggregated energy intensity. Considering the importance of energy efficiency study and the ability of DEA in combining multiple factors, it is reasonable to believe that DEA would play a more important role in energy efficiency studies in future.

In addition to the three application areas discussed, DEA has also been applied to study the productive efficiency of some specific energy sectors, e.g. district heating plants (Raczka, 2001; Agrell and Bogetoft, 2005; Munksgaard et al., 2005), oil and gas industries (Thompson et al., 1995, 1996; Price and Weyman-Jones, 1996; Hawdon, 2003; Kashani, 2005a,b) and coal mines (Byrnes et al., 1984, 1988; Thompson et al., 1992; Kulshreshtha and Parikh, 2002). Besides, as illustrated in Brännlund et al. (1998), DEA is also an effective tool for studying the issue of emissions permit allocations which is an important environmental topic. Among these studies, two recent publications are worth highlighting because they may provide new directions for future research. One was by Ramanathan (2005b) who applied DEA to forecast energy consumption and carbon dioxide emissions. The other was by Pasurka (2006) who established a linkage between DEA and another popular E&E modeling technique called index decomposition analysis (Ang and Zhang, 2000; Ang, 2004).

4.2. Methodological aspect

As shown in Table 1, the methodological aspect is further characterized by the disposability and operating characteristics of inputs and outputs, the RTS property of reference technology, efficiency measure type and the use of MPI as described in Section 3. However, for those studies dealing with DEA models in multiplier form, which take up

² In their excellent survey papers, Tyteca (1996) and Allen (1999) have shown the high potential of DEA in measuring environmental performance in detail.

17% of the 100 studies, we only present the models used without further characterizing their reference technologies and efficiency measures. As a result, these studies are not accounted for in discussing reference technology and efficiency measure.

Not surprisingly, almost all the studies assume that inputs are strongly disposable. This is determined by the “the less the better” property of inputs. However, the studies by Oude Lansink and Bezlepkin (2003) and Oude Lansink and Silva (2003) are two exceptions, in which carbon dioxide is used as an input of greenhouse firms. Several other studies assumed both strong disposability and weak disposability of inputs in order to measure the congestion of inputs, e.g. Byrnes et al. (1984), Byrnes et al. (1988) and Färe et al. (1985).

As to the disposability of outputs, strong disposability is still the most widely used one. Nevertheless, over a quarter of the studies assumed that outputs are weakly disposable, i.e. they dealt with the environmental DEA technology concept. Of these studies, most pertain to such issues as productivity estimation with pollutants considered, environmental performance measurement and estimation of environmental regulation impacts. Examples of such studies include Färe et al. (1986, 1989, 1996), Chung et al. (1997), Hernández-Sancho et al. (2000), Tyteca (1997), Zaim and Taskin (2000a,b), Zofio and Prieto (2001), Zaim (2004) and Zhou et al. (2006b, in press, 2007). A common feature of these studies is that both desirable outputs and undesirable outputs are simultaneously considered. In such cases, the environmental DEA technology is particularly attractive because it has good theoretical properties and could characterize the real production process better.

For the RTS property of reference technology, it is found that about a half of the studies assumed that the reference technology exhibits CRS, although VRS might be a more appropriate assumption (Ramanathan, 2003). One possible reason is that the output-oriented radial efficiency measure is just the reciprocal of the input-oriented radial efficiency measure under the CRS assumption. As a result, the choice between input-oriented and output-oriented DEA model becomes indifferent. In addition, this could be partially explained by the popularity of MPI and the fact that the MPI based on the CRS assumption can be interpreted as a total productivity index (Førsund and Kittelsen, 1998). It is also noted that 30% of the studies adopted both the CRS and the VRS reference technologies. This might be a standardized application mode of DEA

because in such cases the scale efficiency of each DMU can be estimated. Examples of such studies include Raczka (2001), Pacudan and de Guzman (2002), Chien et al. (2003) and Ramanathan (2005). If the CRS, VRS and NIRS reference technologies are used together, the RTS properties of DMUs can be further investigated (Färe et al., 1983, 1985, 1994b; Ramanathan, 2003).

From Table 1 we can find that radial efficiency measure, adopted in over three-quarters of the studies, has all along been the most commonly used one among the various types of efficiency measures. Nevertheless, in some cases other efficiency measures may be more meaningful and practical. For instance, if both desirable and undesirable outputs are considered simultaneously, the DDF efficiency measure may provide a more reasonable productivity index because it considers the output of pollution abatement activities (Chung et al., 1997; Färe et al., 2001; Picazo-Tadeo et al., 2005). Compared with radial efficiency measure, slacks-based efficiency measure provides a more practical index with higher discriminating power for measuring energy efficiency and modeling environmental performance (Hu and Wang, 2006; Hu and Kao, 2007; Zhou et al., 2006b).

In addition, as shown in Table 1, there has been a growing interest on the use of MPI in E&E studies in recent years. Particularly, in 1999–2006, the share taken up by MPI applications is 21%. A majority of MPI applications deal with the study of productivity growth over time in electricity utilities, e.g. Färe et al. (1990), Hjalmarsson and Veiderpass (1992), Yunos and Hawdon (1997), Førsund and Kittelsen (1998), Sueyoshi and Goto (2001), Giannakis et al. (2005), Abbott (2006) and Pombo and Taborda (2006). The popularity of MPI is likely due to the deregulation of electricity sectors worldwide and the interests of regulators in national/international benchmarking of electricity utilities under the new regimes.

4.3. Other features and findings

Table 1 has shown that many studies dealt with both the theoretical and application aspects of DEA. This should attribute to the flexibility and ability of DEA in allowing for varying situations. Since various application studies have their individual characteristics, practitioners and researchers may have to present new DEA versions for their use. Another possible reason is that such popular software packages as EXCEL and MATLAB offer

researchers huge flexibility to construct and apply their own models. In addition, we have also found that many OR/MS researchers favor DEA models in the multiplier form while E&E researchers and economists favor DEA models in the envelopment form. This is likely due to the interdisciplinary nature of DEA and its historical diffusion patterns (Førsund and Sarafoglou, 2005).

A majority of past studies dealt with the input-oriented DEA models rather than the output-oriented ones, which is not specified in Table 1. To a large extent, it should be attributed to the characteristic of energy sectors that higher priority has often been given to the goal of meeting demand (Färe et al., 1994b). As a result, input conservation for given outputs seems to be a reasonable logic. Another possible reason is that in many empirical studies, particularly at the macro level, there is only one output such as GDP but multiple inputs are often used. Nevertheless, the undesirable outputs orientation DEA models seem to be more popular in modeling environmental performance.

5. Model selection and related issues

Since a large number of DEA models are available, researchers using DEA to study E&E issues will inevitably face the problem of deciding a specific DEA version to apply and selecting the appropriate inputs and outputs. Although the general application procedure of DEA has been discussed in previous studies (e.g. Golany and Roll, 1989; Dyson et al., 2001; Ramanathan, 2003), a systematic summary and reconsideration of these guidelines with particular reference to the use of DEA in E&E studies is still meaningful.

In general, input-orientated DEA models might be suitable for most E&E issues in which outputs are decided by the requirement of demand. When undesirable outputs are not considered, the incorporation of radial efficiency measures with the ordinary CRS and VRS reference technologies, e.g. the CCR and BCC models, would be appropriate since such a setting can provide the information on not only technical efficiency but also scale efficiency. If the RTS property of some specific DMU is of interest, the NIRS reference technology can be used as an auxiliary tool to investigate this property. If the performance of DMUs over time is of interest, the nonparametric MPI based on the CCR model is highly recommended because the index obtained can be interpreted as a total factor productivity index.

However, when undesirable outputs are considered, the incorporation of environmental DEA technology with DDF or hyperbolic efficiency measure might be more appropriate for estimating the productive efficiency of DMUs. If a composite economic-energy-environmental index is expected, as illustrated by Zhou et al. (2006b), the slacks-based efficiency measure might be more appropriate because of its higher discriminating power. In addition, the nonparametric MPI can also be integrated with environmental DEA technology for time-series analysis.

As to the selection of inputs and outputs, the first step is to establish a list of possible inputs and outputs that may be related to the study. These inputs and outputs can be further examined by some screening procedures such as preliminary judgment and statistical analysis in order to retain only the most relevant ones.³ Besides, the selection of inputs and outputs also depends on data availability and the number of DMUs. In empirical applications, two widely adopted rules of thumb are to let the number of DMUs larger than the product and be at least two times larger than the sum of the number of inputs and outputs (Dyson et al., 2001; Ramanathan, 2003, 2005a).

Once the DEA models used are specified and the inputs and outputs are determined, the DEA efficiency scores of DMUs can be calculated by some specialized software packages such as DEAP and EMS or by user self-coded programs built upon the EXCEL/MATLAB platforms. These efficiency scores can be further analyzed by using such techniques as regressions analysis in order to obtain more information.

Despite its various strengths, DEA was treated in most application studies as a deterministic technique and the results obtained are therefore very sensitive to even small permutations on the data used. To deal with the issue, Banker (1993) provided a statistical foundation by which hypothesis tests can be carried out based on the DEA efficiency scores. A number of statistical tests have so far been developed to address such issues as comparing the efficiency of two groups of DMUs, examining the existence of scale economics and test the shift of efficiency frontier (Banker, 1996; Kittelsen, 1999;

³ In their application study on evaluating the operating efficiency of a sample of electricity generation plants, Golany et al. (1994) provided an excellent paradigm for the screening process.

Banker and Natarajan, 2004). Another alternative, as suggested by Simar and Wilson (1998), is to use the bootstrap technique to conduct a sensitivity analysis on the DEA efficiency scores. In E&E studies, the usefulness of the bootstrap DEA approach has been empirically demonstrated by Hawdon (2003) and Sanhueza et al. (2004).

6. Conclusions

DEA is becoming more and more popular in E&E studies in recent years. However, there is a lack of a literature review in this field. In this paper we present the results of a survey of DEA in E&E studies which will be useful to researchers entering this exciting field.

Our survey found that benchmarking of electricity utilities accounts for the largest number of studies, which is followed by the areas of modeling environmental performance and energy efficiency study. In the methodological aspect, we found that the CRS reference technology and the radial efficiency measures are still the most widely used specifications. When both desirable outputs and undesirable outputs are considered simultaneously, we have also found that the incorporation of environmental DEA technology with DDF efficiency measure could be a good choice. In addition, there has been a growing interest on the use of nonparametric MPI in E&E studies in recent years.

Based on our survey, we have discussed some issues on the selection of appropriate DEA models, which could be helpful to researchers interested in using DEA to study E&E issues. Given the importance of E&E modeling techniques and the growing interest on performance comparisons, we believe basic DEA models as well as its miscellaneous extensions would likely to play a more important role in E&E studies in future.

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References

Abbott, M., 2005. Determining levels of productivity and efficiency in the electricity industry. *The Electricity Journal* 18 (9), 62–72.

Abbott, M., 2006. The productivity and efficiency of the Australian electricity supply industry. *Energy Economics* 28, 444–454.

Afriat, S., 1972. Efficiency estimation of production functions. *International Economic Review* 13, 568–598.

Agrell, P.J., Bogetoft, P., 2005. Economic and environmental efficiency of district heating plants. *Energy Policy* 33, 1351–1362.

Allen, K., 1999. DEA in the ecological context – an overview. In: Westermann, G. (Ed.), *Measuring the Efficiency in the Private and Public Service Sector*. Gabler, Wiesbaden, pp. 203–235.

Allen, R., Athanassopoulos, A., Dyson, R.G., Thanassoulis, E., 1997. Weights restrictions and value judgements: Evolution, development and future directions. *Annals of Operations Research* 73, 13–34.

Ang, B.W., 2004. Decomposition analysis for policymaking in energy: Which is the preferred method?. *Energy Policy* 32, 1131–1139.

Ang, B.W., 2006. Monitoring changes in economy-wide energy efficiency: From energy-GDP ratio to composite efficiency index. *Energy Policy* 34, 574–582.

Ang, B.W., Zhang, F.Q., 2000. A survey of index decomposition analysis in energy and environmental studies. *Energy* 25, 1149–1176.

Arcelus, F.J., Arocena, P., 2005. Productivity differences across OECD countries in the presence of environmental constraints. *Journal of the Operational Research Society* 56, 1352–1362.

Athanassopoulos, A.D., Lambroukos, N., Seiford, L., 1999. Data envelopment scenario analysis for setting targets to electricity generating plants. *European Journal of Operational Research* 115, 413–428.

Bagdadioglu, N., Price, C.M.W., Weyman-Jones, T.G., 1996. Efficiency and ownership in electricity distribution: A non-parametric model of the Turkish experience. *Energy Economics* 18, 1–23.

Banker, R.D., 1993. Maximum likelihood, consistency and data envelopment analysis: A statistical foundation. *Management Science* 39, 1265–1273.

Banker, R.D., 1996. Hypothesis tests using data envelopment analysis. *Journal of Productivity Analysis* 7, 139–159.

Banker, R.D., Charnes, A., Cooper, W.W., 1984. Some models for estimating technical and scale inefficiencies in data envelopment analysis. *Management Science* 30, 1078–1092.

Banker, R.D., Morey, R.C., 1986a. Efficiency analysis for exogenously fixed inputs and outputs. *Operations Research* 34, 513–521.

Banker, R.D., Morey, R.C., 1986b. The use of categorical variables in data envelopment analysis. *Management Science* 32, 1613–1627.

Banker, R.D., Natarajan, R., 2004. Statistical tests based on DEA efficiency scores. In: Cooper, W.W., Seiford, L., Zhu, J. (Eds.), *Handbook on Data Envelopment Analysis*. Kluwer Academic Publishers, Boston, pp. 299–321.

Barla, P., Perelman, S., 2005. Sulphur emissions and productivity growth in industrialized countries. *Annals of Public and Cooperative Economics* 76, 275–300.

Bevilacqua, M., Braglia, M., 2002. Environmental efficiency analysis for ENI oil refineries. *Journal of Cleaner Production* 10, 85–92.

- Boyd, G.A., McClelland, J.D., 1999. The impact of environmental constraints on productivity improvement in integrated paper plants. *Journal of Environmental Economics and Management* 38, 121–142.
- Boyd, G.A., Pang, J.X., 2000. Estimating the linkage between energy efficiency and productivity. *Energy Policy* 28, 289–296.
- Boyd, G.A., Tolley, G., Pang, J., 2002. Plant level productivity, efficiency, and environmental performance of the container glass industry. *Environmental and Resource Economics* 23, 29–43.
- Brännlund, R., Chung, Y.H., Färe, R., Grosskopf, S., 1998. Emissions trading and profitability: The Swedish pulp and paper industry. *Environmental and Resource Economics* 12, 345–356.
- Byrnes, P., Färe, R., Grosskopf, S., 1984. Measuring productivity efficiency: An application to Illinois strip mines. *Management Science* 30, 671–681.
- Byrnes, P., Färe, R., Grosskopf, S., Lovell, C.A.K., 1988. The effect of unions on productivity: US surface mining of coal. *Management Science* 34, 1037–1053.
- Callens, I., Tyteca, D., 1999. Towards indicators of sustainable development for firms: A productive efficiency perspective. *Ecological Economics* 28, 41–53.
- Caves, D.W., Christensen, L.R., Diewert, W.E., 1982. Multilateral comparisons of output, input and productivity using index numbers. *The Economic Journal* 92, 73–86.
- Charnes, A., Cooper, W.W., Rhodes, E., 1978. Measuring the efficiency of decision making units. *European Journal of Operational Research* 2, 429–444.
- Chauhan, N.S., Mohapatra, P.K.J., Pandey, K.P., 2006. Improving energy productivity in paddy production through benchmarking – an application of data envelopment analysis. *Energy Conversion and Management* 47, 1063–1085.
- Chien, C.F., Lo, F.Y., Lin, J.T., 2003. Using DEA to measure the relative efficiency of the service center and improve operation efficiency through reorganization. *IEEE Transactions on Power Systems* 18, 366–373.
- Chitkara, P., 1999. A data envelopment analysis approach to evaluation of operational inefficiencies in power generating units: A case study of Indian power plants. *IEEE Transactions on Power Systems* 14, 419–425.
- Chung, Y.H., Färe, R., Grosskopf, S., 1997. Productivity and undesirable outputs: A directional distance function approach. *Journal of Environmental Management* 51, 229–240.
- Claggett Jr., E.T., Ferrier, G.D., 1998. The efficiency of TVA power distributors. *Managerial and Decision Economics* 19, 365–376.
- Coelli, T.J., 1996. A Guide to DEAP Version 2.1: A Data Envelopment Analysis (Computer) Program, CEPA Working Paper 96/8, Department of Econometrics, University of New England, Armidale NSW Australia.
- Cook, W.D., Green, R.H., 2005. Evaluating power plant efficiency: A hierarchical model. *Computers & Operations Research* 32, 813–823.
- Cooper, W.W., Seiford, L.M., Tone, T., 2006. *Introduction to Data Envelopment Analysis and Its Uses: With DEA-Solver Software and References*. Springer, New York.
- Cooper, W.W., Seiford, L.M., Zhu, J., 2004. Data envelopment analysis: History, models and interpretations. In: Cooper, L.M., Seiford, L.M., Zhu, J. (Eds.), *Handbook on Data Envelopment Analysis*. Kluwer Academic Publishers, Boston, pp. 1–39.
- Criswell, D.R., Thompson, R.G., 1996. Data envelopment analysis of space and terrestrially-based large scale commercial power systems for earth: A prototype analysis of their relative economic advantages. *Solar Energy* 56, 119–131.
- Dyckhoff, H., Allen, K., 2001. Measuring ecological efficiency with data envelopment analysis (DEA). *European Journal of Operational Research* 132, 312–325.
- Dyson, R.G., Allen, R., Camanho, A.S., Podinovski, V.V., Sarrico, C.S., Shale, E.A., 2001. Pitfalls and protocols in DEA. *European Journal of Operational Research* 132, 242–259.
- Dyson, R.G., Thanassoulis, E., 1988. Reducing weight flexibility in data envelopment analysis. *Journal of the Operational Research Society* 39, 563–576.
- Edvardsen, D.F., Førsund, F.R., 2003. International benchmarking of electricity distribution utilities. *Resource and Energy Economics* 25, 353–371.
- Färe, R., Grosskopf, S., 2004. Modeling undesirable factors in efficiency evaluation: Comment. *European Journal of Operational Research* 157, 242–245.
- Färe, R., Grosskopf, S., Hernandez-Sancho, F., 2004. Environmental performance: An index number approach. *Resource and Energy Economics* 26, 343–352.
- Färe, R., Grosskopf, S., Lindgren, B., Roos, P., 1994a. Productivity developments in Swedish hospitals: A Malmquist output index approach. In: Charnes, A., Cooper, W.W., Lewin, A.Y., Seiford, L.M. (Eds.), *Data Envelopment Analysis: Theory, Methodology and Applications*. Kluwer Academic Publishers, Boston, pp. 253–272.
- Färe, R., Grosskopf, S., Logan, J., 1983. The relative efficiency of Illinois electric utilities. *Resources and Energy* 5, 349–367.
- Färe, R., Grosskopf, S., Logan, J., 1985. The relative performance of publicly-owned and privately-owner electric utilities. *Journal of Public Economics* 26, 89–106.
- Färe, R., Grosskopf, S., Lovell, C.A.K., 1994b. *Production frontiers*. Cambridge University Press, Cambridge.
- Färe, R., Grosskopf, S., Lovell, C.A.K., Pasurka, C., 1989. Multilateral productivity comparisons when some outputs are undesirable: A nonparametric approach. *The Review of Economics and Statistics* 71, 90–98.
- Färe, R., Grosskopf, S., Noh, D., Weber, W., 2005. Characteristics of a polluting technology: Theory and practice. *Journal of Econometrics* 126, 469–492.
- Färe, R., Grosskopf, S., Pasurka, C., 1986. Effects on relative efficiency in electric power generation due to environmental controls. *Resources and Energy* 8, 167–184.
- Färe, R., Grosskopf, S., Pasurka Jr., C.A., 2001. Accounting for air pollution emissions in measures of state manufacturing productivity growth. *Journal of Regional Science* 41, 381–409.
- Färe, R., Grosskopf, S., Roos, P., 1998. Malmquist productivity indexes: A survey of theory and practice. In: Färe, R., Grosskopf, S., Russell, R.R. (Eds.), *Index Numbers: Essays in Honour of Sten Malmquist*. Kluwer Academic Publishers, Boston, pp. 127–190.
- Färe, R., Grosskopf, S., Tyteca, D., 1996. An activity analysis model of the environmental performance of firms – application to fossil-fuel-fired electric utilities. *Ecological Economics* 18, 161–175.

- Färe, R., Grosskopf, S., Yaisawarng, S., Li, S.K., Wang, Z.P., 1990. Productive growth in Illinois electric utilities. *Resources and Energy* 12, 383–398.
- Farrell, M.J., 1957. The measurement of productive efficiency. *Journal of the Royal Statistical Society, Series A (General)* 120, 253–281.
- Ferrier, G.D., Hirschberg, J.G., 1992. Climate control policy. *The Energy Journal* 13, 37–54.
- Førsund, F.R., Kittelsen, S.A.C., 1998. Productivity development of Norwegian electricity distribution utilities. *Resource and Energy Economics* 20, 207–224.
- Førsund, F.R., Sarafoglou, N., 2002. On the origins of data envelopment analysis. *Journal of Productivity Analysis* 17, 23–40.
- Førsund, F.R., Sarafoglou, N., 2005. The tale of two research communities: The diffusion of research on productive efficiency. *International Journal of Production Economics* 98, 17–40.
- Giannakis, D., Jamasb, T., Pollitt, M., 2005. Benchmarking and incentive regulation of quality of service: An application to the UK electricity distribution networks. *Energy Policy* 33, 2256–2271.
- Golany, B., Roll, Y., 1989. An application procedure for DEA. *Omega* 17, 237–250.
- Golany, B., Roll, Y., Rybak, D., 1994. Measuring efficiency of power plants in Israel by data envelopment analysis. *IEEE Transactions on Engineering Management* 41, 291–301.
- Goto, M., Tsutsui, M., 1998. Comparison of productive and cost efficiencies among Japanese and US electric utilities. *Omega* 26, 177–194.
- Hattori, T., Jamasb, T., Pollitt, M., 2005. Electricity distribution in the UK and Japan: A comparative efficiency analysis 1985–1998. *The Energy Journal* 26, 23–47.
- Hawdon, D., 2003. Efficiency, performance and regulation of the international gas industry – a bootstrap DEA approach. *Energy Policy* 31, 1167–1178.
- Haynes, K.E., Ratick, S., Cummings-Saxton, J., 1994. Toward a pollution abatement monitoring policy: Measurements, model mechanics, and data requirements. *The Environmental Professional* 16, 292–303.
- Hernández-Sancho, F., Picazo-Tadeo, A.J., Reig-Martínez, E., 2000. Efficiency and environmental regulation. *Environmental and Resource Economics* 15, 365–378.
- Hjalmarsson, L., Veiderpass, A., 1992. Productivity in Swedish electricity retail distribution. *Scandinavian Journal of Economics* 94 (supplement), 193–205.
- Hu, J.L., Kao, C.H., 2007. Efficient energy-saving targets for APEC economies. *Energy Policy* 35, 373–382.
- Hu, J.L., Wang, S.C., 2006. Total-factor energy efficiency of regions in China. *Energy Policy* 34, 3206–3217.
- Huang, J.P., Poh, K.L., Ang, B.W., 1995. Decision analysis in energy and environmental modeling. *Energy* 20, 843–855.
- Jamasb, T., Nillesen, P., Pollitt, M., 2004. Strategic behaviour under regulatory benchmarking. *Energy Economics* 26, 825–843.
- Jamasb, T., Pollitt, M., 2001. Benchmarking and regulation: International electricity experience. *Utilities Policy* 9, 107–130.
- Jamasb, T., Pollitt, M., 2003. International benchmarking and regulation: An application to European electricity distribution utilities. *Energy Policy* 31, 1609–1622.
- Jebaraj, S., Iniyan, S., 2006. A review of energy models. *Renewable and Sustainable Energy Reviews* 10, 281–311.
- Kashani, H.A., 2005a. State intervention causing inefficiency: An empirical analysis of the Norwegian Continental Shelf. *Energy Policy* 33, 1998–2009.
- Kashani, H.A., 2005b. Regulation and efficiency: An empirical analysis of the United Kingdom continental shelf petroleum industry. *Energy Policy* 33, 915–925.
- Kittelsen, S.A.C., 1999. Mote Carlo simulations of DEA measures and hypothesis tests. Memorandum 9/1999, Department of Economics, University of Oslo.
- Koopmans, T.C., 1951. Analysis of production as an efficient combination of activities. In: Koopmans, T.C. (Ed.), *Activity Analysis of Production and Allocation*. John Wiley, New York.
- Korhonen, P.J., Luptacik, M., 2004. Eco-efficiency analysis of power plants: An extension of data envelopment analysis. *European Journal of Operational Research* 154, 437–446.
- Korhonen, P.J., Syrjänen, M.J., 2003. Evaluation of cost efficiency in Finnish electricity distribution. *Annals of Operations Research* 121, 105–122.
- Kulshreshtha, M., Parikh, J.K., 2002. Study of efficiency and productivity growth in opencast and unsergroud coal mining in India: A DEA analysis. *Energy Economics* 24, 439–453.
- Kumar, S., 2006. Environmentally sensitive productivity growth: A global analysis using Malmquist-Luenberger index. *Eco-logical Economics* 56, 280–293.
- Lam, P.L., Shiu, A., 2001. A data envelopment analysis of the efficiency of China's thermal power generation. *Utilities Policy* 10, 75–83.
- Liang, L., Wu, D., Hua, Z., 2004. MES-DEA modelling for analyzing anti-industrial pollution efficiency and its application in Anhui province of China. *International Journal of Global Energy Issues* 22, 88–98.
- Lo, F.Y., Chien, C.F., Lin, J.T., 2001. A DEA study to evaluate the relative efficiency and investigate the district reorganization of the Taiwan power company. *IEEE Transactions on Power Systems* 16, 170–178.
- Loken, E., 2007. Use of multicriteria decision analysis methods for energy planning problems. *Renewable and Sustainable Energy Reviews* 11, 1584–1595.
- Malmquist, S., 1953. Index numbers and indifference surfaces. *Trabajos de Estadística* 4, 209–242.
- Miliotis, P.A., 1992. Data envelopment analysis applied to electricity distribution districts. *Journal of the Operational Research Society* 43, 549–555.
- Munksgaard, J., Pade, L.L., Fristrup, P., 2005. Efficiency gains in Danish district heating. Is there anything to learn from benchmarking? *Energy Policy* 33 1986–1997.
- Murillo-Zamorano, L.R., 2005. The role of energy in productivity growth: A controversial issue? *The Energy Journal* 26 69–88.
- Nag, B., 2006. Estimation of carbon baselines for power generation in India: The supply side approach. *Energy Policy* 34, 1399–1410.
- Olatubi, W.O., Dismukes, D.E., 2000. A data envelopment analysis of the levels and determinants of coal-fired electric power generation performance. *Utilities Policy* 9, 47–59.
- Onut, S., Soner, S., 2006. Energy efficiency assessment for the Antalya Region hotels in Turkey. *Energy and Buildings* 38, 964–971.
- Oude Lansink, A., Bezlepkin, I., 2003. The effect of heating technologies on CO₂ and energy efficiency of Dutch greenhouse firms. *Journal of Environmental Management* 68, 73–82.

- Oude Lansink, A., Silva, E., 2003. CO₂ and energy efficiency of different heating technologies in the Dutch glasshouse industry. *Environmental and resource economics* 24, 395–407.
- Pacudan, R., de Guzman, E., 2002. Impact of energy efficiency policy to productive efficiency of electricity distribution industry in the Philippines. *Energy Economics* 24, 41–54.
- Pahwa, A., Feng, X., Lubkeman, D., 2002. Performance evaluation of electric distribution utilities based on data envelopment analysis. *IEEE Transactions on Power Systems* 17, 400–405.
- Park, S.U., Lesourd, J.B., 2000. The efficiency of conventional fuel power plants in South Korea: A comparison of parametric and non-parametric approaches. *International Journal of Production Economics* 63, 59–67.
- Pasurka Jr., C.A., 2006. Decomposing electric power plant emissions within a joint production framework. *Energy Economics* 28, 26–43.
- Picazo-Tadeo, A.J., Reig-Martínez, E., Hernández-Sancho, F., 2005. Directional distance functions and environmental regulation. *Resource and Energy Economics* 27, 131–142.
- Podinovski, V.V., 1999. Side effects of absolute weight bounds in DEA models. *European Journal of Operational Research* 115, 583–595.
- Podinovski, V.V., 2001. Validating absolute weight bounds in Data Envelopment Analysis (DEA) models. *Journal of the Operational Research Society* 52, 221–225.
- Pollitt, M.G., 1996. Ownership and efficiency in nuclear power production. *Oxford Economic Papers* 48, 342–360.
- Pombo, C., Taborda, R., 2006. Performance and efficiency in Colombia's power distribution system: Effects of the 1994 reform. *Energy Economics* 28, 339–369.
- Price, C.W., Weyman-Jones, T., 1996. Malmquist indices of productivity change in the UK gas industry before and after privatization. *Applied Economics* 28, 29–39.
- Raczka, J., 2001. Explaining the performance of heat plants in Poland. *Energy Economics* 23, 355–370.
- Ramanathan, R., 2000. A holistic approach to compare energy efficiencies of different transport modes. *Energy Policy* 28, 743–747.
- Ramanathan, R., 2001. Comparative risk assessment of energy supply technologies: A data envelopment analysis approach. *Energy* 26, 197–203.
- Ramanathan, R., 2002. Combining indicators of energy consumption and CO₂ emissions: A cross-country comparison. *International Journal of Global Energy Issues* 17, 214–227.
- Ramanathan, R., 2003. *An Introduction to Data Envelopment Analysis: A Tool for Performance Measurement*. Sage Publications, New Delhi.
- Ramanathan, R., 2005a. An analysis of energy consumption and carbon dioxide emissions in countries of the Middle East and North Africa. *Energy* 30, 2831–2842.
- Ramanathan, R., 2005b. Estimating energy consumption of transport modes in India using DEA and application to energy and environmental policy. *Journal of the Operational Research Society* 56, 732–737.
- Resende, M., 2002. Relative efficiency measurement and prospects for yardstick competition in Brazilian electricity distribution. *Energy Policy* 30, 637–647.
- Sanhueza, R., Rudnick, H., Lagunas, H., 2004. DEA efficiency for the determination of the electric power distribution added value. *IEEE Transactions on Power Systems* 19, 919–925.
- Sarkis, J., Weinrach, J., 2001. Using data envelopment analysis to evaluate environmentally conscious waste treatment technology. *Journal of Cleaner Production* 9, 417–427.
- Scheel, H., 2000. EMS: Efficiency Measurement System User's Manual. Available at: <http://www.wiso.uni-dortmund.de/LSFG/OR/scheel/ems/>.
- Scheel, H., 2001. Undesirable outputs in efficiency valuations. *European Journal of Operational Research* 132, 400–410.
- Seiford, L.M., Thrall, R.M., 1990. Recent developments in DEA: The mathematical programming approach to frontier analysis. *Journal of Econometrics* 46, 7–38.
- Seiford, L.M., Zhu, J., 2002. Modeling undesirable factors in efficiency evaluation. *European Journal of Operational Research* 142, 16–20.
- Shephard, R.W., 1970. *Theory of Cost and Production Functions*. Princeton University Press, Princeton.
- Simar, L., Wilson, P.W., 1998. Sensitivity analysis of efficiency scores: How to bootstrap in nonparametric frontier models. *Management Science* 44, 49–61.
- Sueyoshi, T., 1999. Tariff structure of Japanese electric power companies: An empirical analysis using DEA. *European Journal of Operational Research* 118, 350–374.
- Sueyoshi, T., 2000. Stochastic DEA for restructure strategy: An application to a Japanese petroleum company. *Omega* 28, 385–398.
- Sueyoshi, T., Goto, M., 2001. Slack-adjusted DEA for time series analysis: Performance measurement of Japanese electric power generation industry in 1984–1993. *European Journal of Operational Research* 133, 232–259.
- Stewart, T.J., 1996. Relationships between data envelopment analysis and multicriteria decision analysis. *Journal of the Operational Research Society* 47, 654–665.
- Thakur, T., Deshmukh, S.G., Kaushik, S.C., 2006. Efficiency evaluation of the state owned electric utilities in India. *Energy Policy* 34, 2788–2804.
- Thompson, R.G., Dharmapala, P.S., Thrall, R.M., 1995. Linked-cone DEA profit ratios and technical efficiency with application to Illinois coal mines. *International Journal of Production Economics* 39, 99–115.
- Thompson, R.G., Dharmapala, P.S., Rothenberg, L.J., Thrall, R.M., 1996. DEA/AR efficiency and profitability of 14 major oil companies in US exploration and production. *Computers & Operations Research* 23, 357–373.
- Thompson, R.G., Lee, E., Thrall, R.M., 1992. DEA/AR efficiency of US independent oil/gas producers over time. *Computers & Operations Research* 19, 377–391.
- Tone, 2001. A slacks-based measure of efficiency in data envelopment analysis. *European Journal of Operational Research* 130, 498–509.
- Triantis, K., Otis, P., 2004. Dominance-based measurement of productive and environmental performance for manufacturing. *European Journal of Operational Research* 154, 447–464.
- Tyteca, D., 1996. On the measurement of the environmental performance of firms – a literature review and a productive efficiency perspective. *Journal of Environmental Management* 46, 281–308.
- Tyteca, D., 1997. Linear programming models for the measurement of environmental performance of firms – concepts and empirical results. *Journal of Productivity Analysis* 8, 183–197.

- Tyteca, D., 1998. Sustainability indicators at the firm level: Pollution and resource efficiency as a necessary condition toward sustainability. *Journal of Industrial Ecology* 2, 61–77.
- Von Neumann, J., 1945. A model of general economic equilibrium. *The Review of Economic Studies* 13, 1–9.
- Weyman-Jones, T.G., 1991. Productive efficiency in a regulated industry: The area electricity boards of England and Wales. *Energy Economics* 13, 116–122.
- Yaisawarng, S., Klein, J.D., 1994. The effects of sulfur dioxide controls on productivity change in the US electric power industry. *The Review of Economics and Statistics* 76, 447–460.
- Yunos, J.M., Hawdon, D., 1997. The efficiency of the National Electricity Board in Malaysia: An intercountry comparison using DEA. *Energy Economics* 19, 255–269.
- Zaim, O., 2004. Measuring environmental performance of state manufacturing through changes in pollution intensities: A DEA framework. *Ecological Economics* 48, 37–47.
- Zaim, O., Taskin, F., 2000a. Environmental efficiency in carbon dioxide emissions in the OECD: A non-parametric approach. *Journal of Environmental Management* 58, 95–107.
- Zaim, O., Taskin, F., 2000b. A Kuznets curve in environmental efficiency: An application on OECD countries. *Environmental and Resource Economics* 17, 21–36.
- Zhang, Y., Bartels, B., 1998. The effect of sample size on the mean efficiency in DEA with an application to electricity distribution in Australia, Sweden and New Zealand. *Journal of Productivity Analysis* 9, 187–204.
- Zhou, P., Ang, B.W., Poh, K.L., 2006a. Decision analysis in energy and environmental modeling: An update. *Energy* 31, 2604–2622.
- Zhou, P., Ang, B.W., Poh, K.L., 2006b. Slacks-based efficiency measures for modeling environmental performance. *Ecological Economics* 60, 111–118.
- Zhou, P., Ang, B.W., Poh, K.L., in press. Measuring environmental performance under different environmental DEA technologies. *Energy Economics*.
- Zhou, P., Poh, K.L., Ang, B.W., 2007. A non-radial DEA approach to measuring environmental performance. *European Journal of Operational Research* 178, 1–9.
- Zhu, J., 1996. Data envelopment analysis with preference structure. *Journal of the Operational Research Society* 47, 136–150.
- Zofio, J.L., Prieto, A.M., 2001. Environmental efficiency and regulatory standards: The case of CO₂ emissions from OECD industries. *Resource and Energy Economics* 23, 63–83.